

DATA ANALYTICS PROJECT

UPI Pulse

Digital Payments Growth in India

An interactive dashboard and a 100-record dataset tracking UPI transactions, value, average ticket size, banks live and app share from Apr 2017 to Jul 2025.

100

monthly records

Apr 2017 - Jul 2025

period covered

7

interactive charts

1,967x

monthly volume growth

DEVELOPED BY

Asutosh Desai

September 2026

1 Project overview

The Unified Payments Interface (UPI), built by the National Payments Corporation of India (NPCI), went live in 2016 and has since become India's dominant digital payment rail. **UPI Pulse** is a data analytics project that measures that growth and makes it easy to explore: a cleaned monthly dataset, a set of derived growth metrics, and a bright, responsive dashboard that runs in any browser with no installation.

Objectives

- Quantify how fast UPI transaction volume and value have grown, month by month and fiscal year by fiscal year.
- Show how growth changed over time: rapid early expansion, the COVID-19 dip and recovery, and the recent slowdown in percentage terms.
- Track how the average payment size and the number of participating banks evolved.
- Link movements in the data to policy and product milestones, and show how the app market is split.

Scope at a glance

Item	Detail
Period	Apr 2017 to Jul 2025 (fiscal years FY18 to FY26 year-to-date)
Granularity	Monthly, 100 records
Measures	Transactions (millions), value (Rs crore), average ticket size (Rs), banks live, YoY and MoM growth
Supporting data	Approximate app share for three December snapshots; nine policy and product milestones
Deliverables	CSV dataset, Python build script, interactive HTML dashboard, this report

9.9 mn to 19.47 bn

monthly transactions, Apr 2017 to Jul 2025

Rs 260.6 lakh cr

value transacted in FY25

46 to 675

banks live (approx.)

2 The dataset

The main dataset holds one row per month. It is stored as a CSV (easy to open in Excel, Power BI or pandas) and is also embedded in the dashboard.

Files

File	Purpose
data/upi_monthly.csv	Main dataset: 100 monthly records, 11 columns
data/upi_app_share.csv	Approximate share of UPI volume by app (Dec 2022, Dec 2023, Dec 2024)
dashboard/data.js	Same data as a JavaScript object so the dashboard opens with a double-click
build_dataset.py	Builds and validates both CSV files and regenerates data.js

Columns in upi_monthly.csv

Column	Meaning
month	First day of the month (YYYY-MM-01)
fiscal_year	Indian fiscal year, April to March (e.g. FY25 = Apr 2024 to Mar 2025)
volume_mn	Transactions in the month, in millions
value_cr	Value transacted in the month, in Rs crore (1 lakh crore = 100,000 crore)
avg_ticket_inr	Value divided by volume: average Rs per transaction
banks_live	Banks live on UPI (approximate)
volume_yoy_pct , value_yoy_pct	Growth versus the same month a year earlier
volume_mom_pct , value_mom_pct	Growth versus the previous month
milestone	Policy or product event in that month, if any

How the data was prepared

- Monthly volumes are rounded reference figures based on NPCI's published UPI statistics.
- Each fiscal year is calibrated so its months add up to NPCI's reported annual volume and value. The build script prints every calibration factor; all are within 2.5%.
- Monthly value is computed as volume multiplied by an interpolated average ticket size, then calibrated to the annual value total.
- Growth columns (YoY, MoM) and the average ticket size are derived fields, calculated in build_dataset.py .

Data note. These figures are approximate and were not downloaded live from NPCI. Individual months, and especially monthly value, banks-live and app-share figures, carry more uncertainty than the annual totals. Verify against NPCI's official statistics (npci.org.in, UPI product statistics) before using the numbers outside this project.

3 Key findings

All figures below are calculated from the dataset and update automatically if the data changes.

Fiscal year	Transactions (bn)	Value (Rs lakh crore)	Avg ticket (Rs)	Volume growth vs prior FY
FY18	0.91	1.09	1,191	n/a
FY19	5.35	8.77	1,638	+485.1%
FY20	12.52	21.31	1,702	+133.9%
FY21	22.33	41.04	1,838	+78.4%
FY22	46.00	84.17	1,830	+106.0%
FY23	83.75	139.15	1,661	+82.1%
FY24	131.13	199.89	1,524	+56.6%
FY25	185.85	260.56	1,402	+41.7%
FY26 (4 months)	74.44	97.65	1,312	n/a

What the data shows

- **Explosive scale-up.** Monthly transactions rose from 9.9 million in Apr 2017 to 19.47 billion in Jul 2025, roughly 1,967 times higher.
- **FY25 in numbers.** 185.85 billion transactions worth Rs 260.6 lakh crore, up 41.7% in volume on FY24.
- **Growth is maturing.** Year-on-year growth peaked near +360% in Mar 2019 and has eased to about +35% by Jul 2025: still strong, but on a far larger base.
- **COVID-19 dip and rebound.** Volume fell 19.8% month-on-month in Apr 2020 to 1,000 million after the nationwide lockdown, then recovered within two months.
- **Smaller payments over time.** The average ticket peaked at about Rs 1,878 in Oct 2020 and has drifted down to Rs 1,288, consistent with UPI moving into everyday, lower-value payments.
- **A wider network.** Banks live grew from about 46 to about 675 (approximate).
- **Steady momentum.** Only 26 of 99 months recorded a decline in volume versus the previous month.
- **Concentrated apps.** In the December 2024 snapshot two apps carry roughly 85% of transactions (approximate): PhonePe about 48% and Google Pay about 37%.

4 The dashboard

The dashboard is a single-page web application. Every chart, KPI card, insight and the data table re-render together for the period and measure you select.

Filters

Period presets (all time, last 5 years, last 3 years, last 12 months, COVID era), custom from/to months, a Transactions or Value measure, and a linear or log scale for the trend chart.

KPI cards

Latest-month transactions, value, average ticket size, banks live, and the total for the period, each with a sparkline and change indicator.

Seven charts

Monthly trend with milestone markers, year-on-year growth, average ticket size, fiscal-year totals, banks live, a month-over-month heatmap and app share.

Live insights and table

Plain-language takeaways calculated for the selected period, plus a searchable, sortable table with CSV export of the current view.

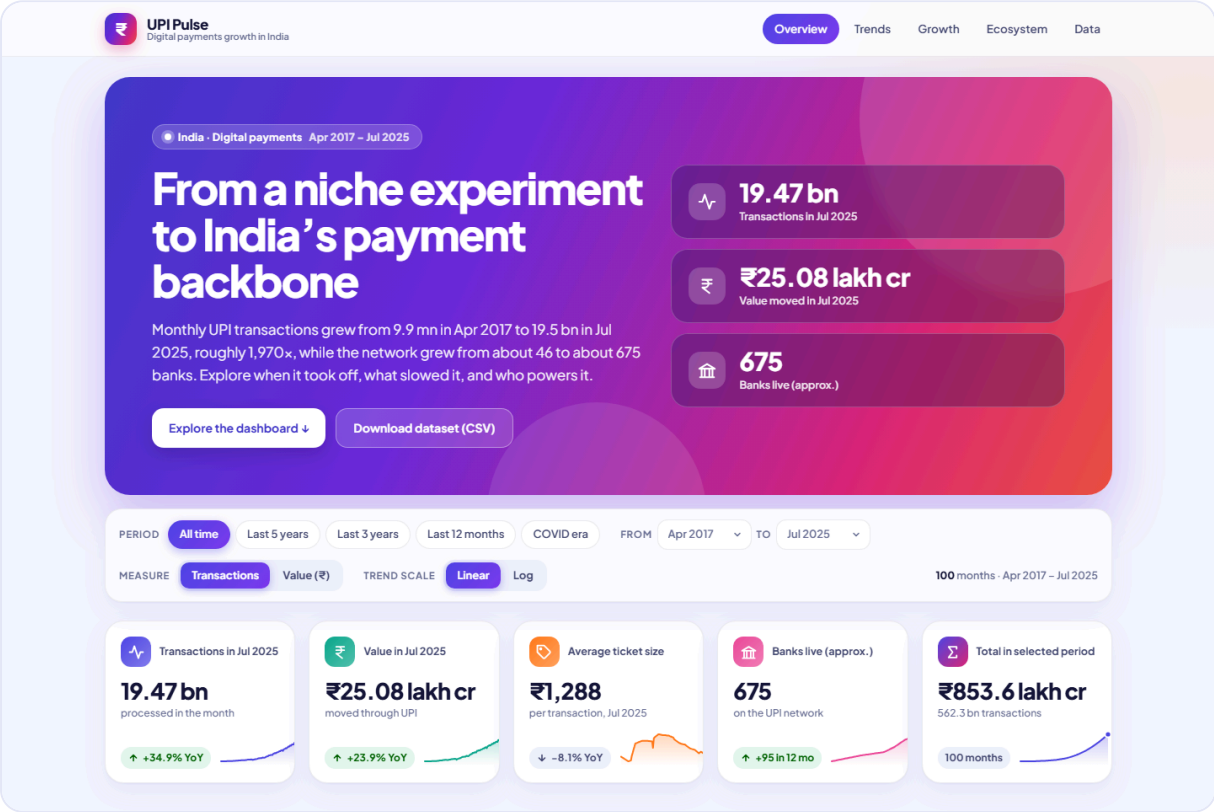


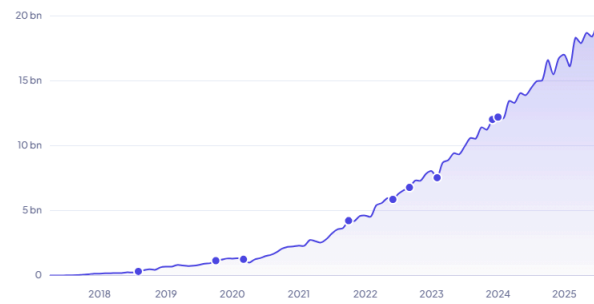
Figure 1. Overview, filters and KPI cards

How fast did UPI scale?

Monthly activity for the selected period. Hover any month for exact values; dots mark policy and product milestones.

Monthly UPI transactions

Apr 2017 – Jul 2025 - linear scale



● Milestone month — hover for details

What the numbers say

Calculated live for your selected period

1,967x growth in monthly transactions
9.9 mn in Apr 2017 to 19.47 bn in Jul 2025 — about 151% a year, annualised.

Dec 2018: +42.9% in a single month
The steepest one-month jump in this period, counting only months that started above 100 mn transactions.

Average payment held steady near ₹1,288
₹1,292 in Apr 2017 to ₹1,288 in Jul 2025.

26 of 99 months dipped
Month-on-month declines were most common in Apr (5 times); growth otherwise held on.

Year-on-year growth in transactions

Growth vs the same month a year earlier — slowing in % terms as the base explodes



Shown only where the same month a year earlier had ≥100 mn transactions.

Average ticket size

Value per transaction (₹) — climbed to a 2020-21 peak, then drifted lower



Figure 2. Trend chart, live insights, year-on-year growth and average ticket size

GROWTH

Year by year, month by month

Annual totals by fiscal year (April–March), banks on the network, and a calendar of month-over-month momentum.

Transactions by fiscal year

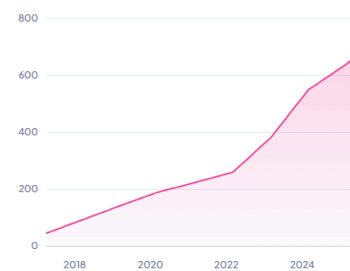
Sum of the months inside your selected period



Lighter bars are partial years: only part of the fiscal year falls in your period, or is reported so far.

Banks live on UPI

Approximate count of participating banks



Month-over-month momentum in transactions

Each cell is growth vs the previous month. Rows are fiscal years, columns run April to March.

	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
FY18	-	-10.0%	+11.1%	+50.0%	+13.3%	+82.4%	+145%	+36.8%	+40.4%	+4.1%	+13.2%	+3.5%
FY19	+9.4%	-0.5%	+30.2%	-7.3%	+36.8%	+29.8%	+19.0%	-10.0%	+42.9%	+8.4%	+0.3%	+18.5%
FY20	-4.1%	-6.1%	+2.9%	+9.0%	+11.7%	+4.0%	+20.2%	+6.1%	+7.4%	-0.2%	+1.5%	-5.9%
FY21	-19.8%	+23.5%	+8.3%	+12.1%	+6.9%	+12.5%	+15.1%	+6.7%	+1.1%	+3.0%	-0.4%	+19.2%
FY22	-3.3%	-3.9%	+10.6%	+15.7%	+9.4%	+2.9%	+15.5%	-0.8%	+9.1%	+1.1%	-1.9%	+19.4%
FY23	+3.2%	+6.6%	-1.5%	+7.3%	+4.5%	+3.1%	+7.8%	+0.0%	+7.2%	+2.7%	-6.3%	+15.3%
FY24	+2.3%	+5.9%	-0.8%	+6.7%	+6.2%	-0.3%	+8.1%	-1.5%	+7.0%	+1.5%	-0.8%	+11.0%
FY25	-1.0%	+5.5%	-1.1%	+4.0%	+3.6%	+0.5%	+10.3%	-6.6%	+8.1%	+1.6%	-5.2%	+13.6%
FY26	-2.2%	+4.4%	-1.5%	+5.8%								

Decline — Growth — colours saturate at ±20%; hover for the exact figure

Figure 3. Fiscal-year totals, banks live and the month-over-month momentum heatmap

Design and accessibility

- Bright, modern palette (indigo, pink, orange, teal) validated for colour-blind readers; each measure keeps the same colour throughout.
- Responsive layout that stacks cleanly on tablets and phones; reduced-motion preferences are respected.
- Hover tooltips with crosshair on every chart, and the full data table acts as an accessible alternative to the charts.
- Year-on-year figures are shown only where the prior-year base was at least 100 million transactions, to avoid misleading percentages.

5 Technology and project structure

Layer	Technology
Data preparation	Python 3 (standard library only): <code>build_dataset.py</code>
Dashboard	HTML5, CSS3 and vanilla JavaScript
Charts	Chart.js 4.4.3, bundled locally in <code>dashboard/vendor/</code> (works offline)
Data format	CSV plus an embedded JavaScript data file

```
UPI & Digital Payments Growth in India/
|-- README.md
|-- build_dataset.py      build + validate the dataset
|-- data/
|   |-- upi_monthly.csv   100 monthly records
|   `-- upi_app_share.csv approximate app share
`-- dashboard/
    |-- index.html        open this file
    |-- styles.css
    |-- app.js             filters, KPIs, charts, table
    |-- data.js            generated data
    `-- vendor/chart.umd.min.js
```

6 How to run the project

Requirements

Requirement	Details
Browser	Any modern browser: Microsoft Edge, Google Chrome, Firefox or Safari
Python 3.8 or newer	Only needed if you want to rebuild the dataset. No extra packages are required.
Internet	Optional. Charts and data work fully offline; only the web font is fetched online (a system font is used otherwise).

1 Open the project folder

Open PowerShell (or Command Prompt) and move into the project folder:

```
cd "E:\DA Project-1\UPI & Digital Payments Growth in India"
```

2 Check Python optional

Only required for step 3.

```
python --version
```

3 Rebuild the dataset optional

The CSV files and `dashboard/data.js` are already included. Run this to regenerate and re-validate them:

```
python build_dataset.py
```

Expected output ends with the record count and fiscal-year totals:

```
Wrote 100 monthly rows: 2017-04-01 -> 2025-07-01
FY18:      915 mn txns   Rs    1.09 lakh cr
...
FY25:   185,850 mn txns   Rs   260.56 lakh cr
```

4

Launch the dashboard

required

Either double-click `dashboard\index.html` , or use a command:

```
# Windows PowerShell
Start-Process ".\dashboard\index.html"

# Windows Command Prompt
start dashboard\index.html

# macOS
open dashboard/index.html

# Linux
xdg-open dashboard/index.html
```

5

Serve it locally

optional

Not required, but useful if you prefer to view the dashboard over a local web address. Run this from the project folder (so the dataset download link also works):

```
python -m http.server 8000
```

Then browse to `http://localhost:8000/dashboard/` . Press `Ctrl + C` in the terminal to stop the server.

Using the dashboard

- Pick a period with the chips, or choose exact **From** and **To** months.
- Switch between **Transactions** and **Value**; turn on **Log** scale to see early growth clearly.
- Hover any chart for exact values; dots on the trend line mark milestones.
- In the **Data** section, search, sort by any column, and use **Export CSV** to save the current view.

Troubleshooting

Problem	Fix
Page says the data or chart library could not load	Run <code>python build_dataset.py</code> , and confirm <code>dashboard/data.js</code> and <code>dashboard/vendor/chart.umd.min.js</code> exist.
<code>python</code> is not recognised	Install Python from python.org (tick "Add to PATH"), or try <code>py build_dataset.py</code> .
Fonts look different offline	Expected: the dashboard falls back to your system font. Data and charts are unaffected.
Export CSV does not download	Open the dashboard in a normal browser window (not an embedded preview) and allow downloads.

7 Updating with official figures

To replace the approximate values with official NPCI statistics, edit the lists at the top of `build_dataset.py`, then rebuild and refresh the browser:

Name in the script	What to change
<code>VOLUME_MN</code>	Monthly transaction volume in millions, starting Apr 2017
<code>FY_TOTAL_VOLUME_MN</code> , <code>FY_TOTAL_VALUE_CR</code>	Official annual totals used to calibrate the months
<code>TICKET_ANCHORS</code>	Known average ticket sizes; months in between are interpolated
<code>BANK_ANCHORS</code>	Known banks-live counts; months in between are interpolated
<code>EVENTS</code>	Milestones shown on the trend chart and timeline
<code>APP_SHARE</code>	App share snapshots (each snapshot must total 100)

```
python build_dataset.py
```

The script checks that there are exactly 100 months. If you add months, update `START_YEAR` / `START_MONTH` as needed and adjust the `assert n == 100` line in `main()`.

UPI Pulse, Digital Payments Growth in India. Developed by **Asutosh Desai**.